Two-Stage Retrieval for RAG on the GDPR: Bi-Encoder Baseline vs. Cross-Encoder Re-Ranking

Code: https://github.com/mahesh006/LMA_ASSIGNMENT

1 Method

1.1 Document and Preprocessing

I used the EU General Data Protection Regulation (Regulation (EU) 2016/679), the official 88-page PDF from EUR-Lex with 54,932 words. It has 173 recitals followed by 99 articles. Many articles state a general rule and then restrict it somewhere else ("shall not apply", "shall be prohibited", "not below 13 years"). Most rules are also explained a second time in a recital, using different wording. Both properties make it a hard document for retrieval.

Text is extracted page by page with `pypdf`. Before chunking I applied three cleaning steps:

- drop the running page header (`Official Journal of the European Union L 119/<n>`);
- re-join words that were hyphenated at a line break;
- collapse all whitespace.

The text is then split into fixed windows of 150 words with a 30-word overlap, which gives **458 chunks**. The longest chunk is 220 tokens, so every chunk fits inside the bi-encoder's 256-token limit and nothing is truncated during embedding.

1.2 Retrieval Pipelines and Architecture Comparison

Both pipelines use the same chunks, the same index and the same top-10 candidate list (Table 1).

Pipeline 1 (Baseline). Every chunk is embedded once with the bi-encoder, and the vectors are stored in a FAISS flat inner-product index. The vectors are L2-normalised, so the inner product is the cosine similarity. For each query the top-10 chunks are retrieved, and the first three go to the LLM.

Pipeline 2 (Re-Ranking). The same ten candidates are scored by the cross-encoder as (query, chunk) pairs and sorted by that score, and the new top three go to the LLM.

Table 1: Experimental setup.

Bi-encoder	`sentence-transformers/all-MiniLM-L6-v2` (384-d), FAISS `IndexFlatIP`
Cross-encoder	`cross-encoder/ms-marco-MiniLM-L-6-v2`
LLM	`Qwen/Qwen2.5-7B-Instruct`, float16, greedy decoding, max 250 new tokens
Retrieval	$k = 10$ candidates, $n = 3$ chunks to the LLM
Hardware	4 × NVIDIA RTX 2080 Ti (11 GB), LLM split across GPUs with `device_map="auto"`

The two models differ in *when* the query and the chunk meet (Table 2). The bi-encoder turns each text into a vector independently, so relevance can only be measured as the distance between two fixed summaries. The cross-encoder reads the query and the chunk as one sequence, so attention runs between every query token and every chunk token before it gives a score.

Table 2: Bi-encoder vs. cross-encoder.

	Bi-Encoder	**Cross-Encoder**
Input	Query and chunk encoded separately	`[CLS] query [SEP] chunk [SEP]` as one input
Interaction	Only through the final cosine similarity	Self-attention across all query and chunk tokens
Output	One 384-d vector per text	One relevance score per pair
Pre-compute	All 458 chunk vectors, once, offline	Nothing; each pair is scored at query time
Cost per query	1 encoder pass + index search	N encoder passes for N candidates
Weakness	Ranks by topic; blurs negation and exceptions	Too slow to run over the whole corpus

1.3 Generation

One prompt template and one generation function serve both pipelines. The prompt tells the LLM to answer only from the given chunks, to cite the chunk numbers, and to reply "The provided context does not contain the answer." if the chunks are not enough. I added this rule because Qwen has probably seen the GDPR during training and could otherwise answer from memory, which would hide a retrieval failure. Decoding is greedy, so the only thing that differs between Answer 1 and Answer 2 is the three chunks. Two separate runs produced identical rankings and identical answers.

1.4 Evaluation Protocol

I wrote ten queries covering the failure types in the assignment: keyword traps, exceptions and corner cases, and negation. Each query has one or more *gold phrases*: short verbatim passages of the GDPR that contain the answer. Any chunk containing a gold phrase is a gold chunk, and Table 3 reports the best gold rank in each pipeline. Matching ignores case and whitespace. In my first run some labels were too narrow: for one query the phrase matched an unrelated article, and for another the answer was split across two chunks. Gold sets therefore include both the Article and the matching Recital where they state the same rule. Answers were judged by hand against the gold text: ✓ correct, ∼ partially correct (incomplete or self-contradicting), × wrong or no answer. Latency was measured with `time.perf_counter` after a warm-up call. Retrieval and re-ranking times are averaged over 10 queries × 20 repeats, and generation over the 10 queries.

2 Results

2.1 Rank Shifts

With re-ranking, the gold chunk was in the top three for **10/10** queries, against **7/10** for the baseline (Table 3). Three queries were restored: their gold chunk sat at ranks 4–6 for the bi-encoder and was moved into the top three, twice to rank #1. In those three cases the baseline's #1 chunk matched the topic but not the question:

- Q6: general text about erasure, not the article that lists its exceptions;
- Q7: general text on public-interest tasks, not the sentence about legitimate interests;
- Q8: the exceptions to the duty to *inform* the data subject, not the exceptions to the health-data prohibition.

In the five queries where the baseline already had the gold chunk at #1, re-ranking kept it in the top three.

Table 4 maps these results onto the failure categories named in the assignment.

2.2 Answer Quality

Correct answers rose from 4/10 to 6/10, and wrong answers fell from 4 to 1 (Table 5). Unlike a pure refusal, the baseline often produced a confident answer built from the wrong chunk.

- Q8: it listed the exceptions to the duty to inform as if they were the cases where health data may be processed.
- Q5: it added a "public communication" requirement that belongs to a different condition of Art. 34(3).

Both answers sound plausible and cite a real chunk, which makes them harder to spot than an abstention. After re-ranking, both became correct. In the terms of the assignment:

Table 3: Best gold-chunk rank before (bi-encoder) and after (cross-encoder) re-ranking.

Category	Query	Gold	Bi	CE	Baseline #1 (distractor)
Exception	Q6: When can a request for erasure be refused?	259	#6	#3	c76 general erasure text
Exception	Q7: Can a public authority rely on legitimate interests?	52	#4	#1	c48 public-interest tasks
Negation	Q8: When is processing health data *not* prohibited?	231/61	#5	#1	c73 duty-to-inform exceptions
Keyword trap	Q3: Can a bank reject a loan using only an algorithm?	85	#2	#1	c79 erasure recital
Exception	Q4: Must a 50-employee business keep processing records?	12	#2	#1	c101 processor contracts
Keyword trap	Q1: Maximum penalty for breaking the GDPR?	436	#1	#1	gold already #1
Keyword trap	Q2: Company "hacked": how fast to tell the regulator?	294/105	#1	#1	gold already #1
Exception	Q5: Stolen data was encrypted: must people be told?	297	#1	#1	gold already #1
Negation	Q9: Can a company fire its DPO over its advice?	311	#1	#1	gold already #1
Negation	Q10: Can a 12-year-old consent alone?	229	#1	#1	gold already #1
Gold chunk in top 3			**7/10**	**10/10**	

Bi and CE give the best rank of any gold chunk. Where two chunks are listed, the first is the best gold chunk for the baseline and the second for the re-ranker. In Q8, Art. 9(2) (c231) stayed at #5, while Recital 52 (c61) rose from #7 to #1. In Q2, c294 went from #1 to #3 and c105 from #2 to #1.

Table 4: Failure categories from the assignment and where they are demonstrated.

Category	Evidence	Outcome
Keyword trap / distractor	The baseline #1 chunk in Q6, Q7 and Q8 shares the query's words but answers a different question (c76 erasure in general, c48 public-interest tasks, c73 exceptions to the duty to inform). My own keyword-trap queries Q1–Q3 did *not* fool the bi-encoder.	Distractors demoted (#1 → #8, #4, #4); gold promoted
Exceptions & corner cases	Q6: exceptions to erasure (Art. 17(3)); Q7: public authorities excluded from legitimate interests (Recital 47)	#6 → #3; #4 → #1
Negation & constraints	Q8: when health data is *not* prohibited (Recital 52)	c61 #7 → #1
Buried chunk (ranks 4–10)	Q6, Q7, Q8 above; Q4: Art. 30(5) at #9	Three promoted into the top 3; Q4 only to #4

- Q8 is a *misstated rule*;
- Q5 is a *hallucinated* extra condition;
- Q7 is a missing answer, because the critical chunk sat outside the top 3;
- Q6 *omits a clause* in both pipelines;
- Q4 is a *misstated rule* in both pipelines, with an invented comparison ("50 exceeds 250") in Answer 1.

Better retrieval did not always mean a better answer. In Q6 the baseline answer came from Recital 65 and listed four of the five exceptions in Art. 17(3), missing only freedom of expression. The re-ranked answer included freedom of expression but dropped three of the others, so both are partial. In Q7, Answer 2 states the correct restriction but opens with the refusal sentence from the prompt. I judged it partial because it contradicts itself.

Table 5: Answer 1 (baseline) vs. Answer 2 (re-ranked).

Query	Answer 1	Answer 2	A1	A2
Q8 Health data	Lists "laid down by law, disproportionate effort, archiving" from the wrong rule (c73)	Public interest, health security, control of communicable diseases (Recital 52)	×	✓
Q5 Encrypted data	Not required, but adds a "public communication" duty from Art. 34(3)(c)	Not required if the data are unintelligible, e.g. encrypted	×	✓
Q7 Public authority	"The provided context does not contain the answer."	Restriction in Recital 47, but opens with the refusal sentence	×	~
Q6 Erasure refused	Four exceptions; freedom of expression missing	Freedom of expression, legal obligation, public interest; others missing	~	~
Q3 Loan by algorithm	Allowed only if authorised by law (Recital 71)	Same	~	~
Q4 Records, 50 staff	"50 exceeds the threshold of 250", so records required	Records required	×	×
Q1, Q2, Q9, Q10	Correct (fine cap, 72 hours, DPO protection, age 13–16)	Same	✓	✓
Correct / partial / wrong (out of 10)			**4/2/4**	**6/3/1**

2.3 Latency

Re-ranking ten pairs adds about 16 ms, raising retrieval from 8.0 ms to 24.2 ms (Table 6). That is about 0.4% of the time the LLM needs to write an answer. Generation dominates end-to-end latency and depends mainly on answer length. The re-ranked answers happened to be slightly shorter, so Pipeline 2's generation time is a little lower.

Table 6: Mean latency per query. The baseline has no re-ranking step.

Stage	Pipeline 1	Pipeline 2
Query embedding + FAISS search (ms)	8.02	8.02
Cross-encoder, 10 pairs (ms)	not used	16.17
Retrieval total (ms)	**8.02**	**24.19**
LLM generation (s)	4.38	3.86

3 Discussion

3.1 Why Re-Ranking Helps

All three restored queries fail in the same way: the bi-encoder ranks by shared topic. A recital about erasure, about public-interest tasks or about exceptions in general lies close to the query in embedding space, even though it answers a different question. Words that decide the meaning in legal text, such as "not", "shall not apply" or "by public authorities", change a mean-pooled vector very little. In Q7 the decisive sentence is one line inside a recital that is mostly about something else, so its signal is diluted in the chunk vector. The

cross-encoder reads the question and the chunk together, so it can check whether the chunk actually answers the question rather than just sharing its vocabulary.

3.2 Where Re-Ranking Does Not Help

The cross-encoder can only reorder the ten candidates it receives, and the top-3 cut-off is still tight (Table 7). In Q4 the operative exemption, Art. 30(5), rose from #9 to #4 but never reached the LLM. The only gold chunk in the top three was Recital 13, which mentions the 250-employee derogation without stating the rule. Both answers concluded that the business must keep records, and Answer 1 even claimed "50 exceeds 250". In Q3, Art. 22 rose from #7 to #4, and both answers relied on Recital 71 alone. Re-ranking also does not fix how the LLM uses the context: in Q6 the right article reached the LLM, but the answer was still incomplete.

Table 7: Cases where re-ranking was not enough.

Type	Query	Chunk	Bi → CE	Result
Cut-off too tight	Q4: Must a 50-employee business keep records?	290 (Art. 30(5))	#9 → #4	Both answers wrong
Cut-off too tight	Q3: Can a bank reject a loan by algorithm?	267 (Art. 22)	#7 → #4	Both partial
Generation, not retrieval	Q6: When can erasure be refused?	259 (Art. 17(3))	#6 → #3	Both partial
Self-contradiction	Q7: Public authority, legitimate interests?	52 (Recital 47)	#4 → #1	Correct fact after a refusal line

3.3 Scaling

Running the cross-encoder over the whole corpus would make every query cost one transformer pass per chunk. Its score depends on the query, so none of this work can be done in advance. The bi-encoder computes the chunk vectors once offline, and at query time it needs only one encoding plus a vector search, which approximate indexes such as HNSW keep fast for millions of vectors. I measured the cross-encoder on all 458 chunks for one query: 348.6 ms, against 23.0 ms for the two-stage pipeline, which is about 0.76 ms per pair. Table 8 extrapolates this linearly. The two-stage design spends the cheap model on recall over the whole collection and the expensive model on precision over only $k = 10$ candidates.

Table 8: Per-query cost of cross-encoding the whole corpus (extrapolated from 0.76 ms/pair).

Corpus size	Cross-encoder on all chunks
458 chunks (this document, measured)	0.35 s
10k chunks	∼7.6 s
100k chunks	∼1.3 min
1M chunks	∼12.7 min

3.4 Limitations

- Small evaluation: ten hand-written queries on one document. They show failure modes, not average accuracy.
- Gold labels and answer correctness were judged by me. The gold sets were widened once, to include matching recitals, after the first run.
- Fixed 150-word windows ignore article boundaries, so some rules (e.g. Art. 9(2)) are split across chunks. Chunking by article might help.
- Only $k = 10$ and $n = 3$ were tested. A larger n would likely have fixed Q4 and Q3.
- Latency was measured on one machine with short answers. Only the 458-chunk point of the scaling table is measured; the rest is extrapolated.